

PART-1:

Models used:

google-t5/t5-small base model, ubikpt/t5-small-finetuned-cnn fine-tuned on cnn_dailymail dataset. It had 15,599 articles and summaries. There were no missing values.

Evaluation metrics:

Final evaluation summary					
Model	ROUGE-1	ROUGE-2	ROUGE-L	Perplexity	BERT F1
google-t5/t5-small	0.3427	0.1497	0.3000	54.49	0.8643
ubikpt-finetuned-cnn	0.2906	0.1343	0.2639	98.00	0.8689

BERTSCORE Summary-

Model 1: {'Precision': np.float64(0.8737117948727131), 'Recall': np.float64(0.8553282154340822)}

Model 2: {'Precision': np.float64(0.8870902665494856), 'Recall': np.float64(0.8518118597550365)}

Comments:

- The base T5-small model actually scored higher ROUGE values than the CNN-finetuned model, which was unexpected. Normally, I'd expect the fine-tuned model to do better. This could be due to differences in tokenization or data formatting.
- For Perplexity the base model's average 54.49 was better (lower) than the fine-tuned CNN model 98.00, showing it was more confident in its text generation.
- For BERTScore I report Precision, Recall, and F1 as required. The precision values show how much of the generated summary was relevant 0.8737 for base, 0.8871 for fine-tuned. The Recall values show how much of the

reference summary was captured 0.8553 for base, 0.8518 for fine-tuned. The F1 scores 0.8643 and 0.8689 were very close, showing both models performed similarly in semantic matching.

Qualitative Comparison of Summaries:

Article 4-

Original article (first 500 characters):

A motorist was caught drink-driving - while parking his car near a court where he was due to stand trial. John Graham, 58, left his Toyota MR2 illegally near Plymouth Magistrates' Court, where he was supposed to face a jury on charges of fraud. But a passing police officer spotted the vehicle obstructing an entrance and Graham failed a breath test. John Graham, 58, was caught drink-driving parking his car near a court where he was due to stand trial . He then appeared in two courts on the same d...

Ground truth summary:

John Graham, 58, left his car illegally near Plymouth Magistrates' Court . He was supposed to face a jury on charges on fraud . But a police officer spotted the vehicle and Graham failed a breath test .

Model 1 (google-t5/t5-small) summary:

the 58-year-old left his Toyota MR2 illegally near a court. he was supposed to face a jury on charges of fraud. but passing police officer spotted the vehicle obstructing an entrance. he failed a breath test and appeared in two courts on

the same day.

Model 2 (ubikpt/t5-small-finetuned-cnn) summary:

John Graham, 58, left his Toyota MR2 illegally near Plymouth Magistrates' Court . He was supposed to face a jury on charges of fraud

Comment:

The CNN model was more concise and clean, but the base model included a more complete sequence of events. Between the two, I would say the base T5 actually gave a fuller, more coherent summary.

Article 6-

Original article (first 500 characters):

(CNN Student News) -- December 14, 2012 . Media Literacy Question of the Day . How might loud commercials affect TV viewers' reactions to products being advertised? Do you think that restrictions on commercials' volume will affect viewers and businesses? Explain. * . * . Know Your News -- The following questions relate to events that were covered this week on CNN Student News. Write your answers in the space provided. Click here for a PDF version of this Newsquiz. 1. Johnny Manziel is the first ...

Ground truth summary:

Use the weekly Newsquiz to test your knowledge of stories you saw on CNN Student News . Write your answers in the space provided . Today's Newsquiz includes the Media Literacy Question of the Day .

Model 1 (google-t5/t5-small) summary:

the following questions relate to events covered this week on CNN Student News. write your answers in the space provided.

Model 2 (ubikpt/t5-small-finetuned-cnn) summary:

CNN Student News: How might commercials affect TV viewers' reactions to products being advertised? Write your answers in the space provided

Comment:

Both models stayed close to the quiz theme. The fine-tuned version sounded more

natural and closer to a student prompt, while the base model was more direct and factual.

Summary:

Despite the name, the CNN-finetuned model didn't always outperform the base model. The base T5 seemed to generalize better, while the fine-tuned version produced shorter but slightly less complete summaries.

PART-2:

I fine-tuned the base google-t5/t5-small model on the Aeon Essays dataset. It had 2,235 essays, each paired with a short "description" summary.

Data Split

- Train: 1600
- Validation: 200
- Test: 435

```
Configuring training parameters...
Starting fine-tuning for 3 epochs...
```

[1200/1200 06:17, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Rouge1	Rouge2	RougeL
1	0.844400	0.783637	0.058998	0.006797	0.045867
2	0.791800	0.771402	0.168555	0.019553	0.132750
3	0.788100	0.768957	0.168606	0.018449	0.133543

Final Evaluation:

Evaluation summary:

Model	ROUGE-1	ROUGE-2	ROUGE-L	Perplexity	BERT F1
Original (T5-small)	0.1384	0.0130	0.1039	60.97	0.8390
Fine-tuned (3 Epochs)	0.1522	0.0184	0.1201	92.43	0.8479

Observations

- The metrics show clear improvements in ROUGE scores after fine-tuning:
 - ROUGE-1 increased from 0.1384 to 0.1522 (10.0% improvement)
 - ROUGE-2 increased from 0.0130 to 0.0184 (41.5% improvement)
 - ROUGE-L increased from 0.1039 to 0.1201 (15.6% improvement)

- The ROUGE-2 improvement is particularly significant because it measures bigram overlap, showing the fine-tuned model learned better phrase-level patterns from the essay dataset.
- However, perplexity increased from 60.97 to 92.43. This doesn't necessarily mean the model got worse. Higher perplexity can occur when a model becomes more creative and diverse in its word choices rather than sticking to safe, common words. For essay summarization, where more expressive language is valued, this trade-off might be acceptable.
- BERTScore F1 improved from 0.8390 to 0.8479 indicating the fine-tuned summaries were semantically closer to the reference summaries. While this is a small improvement, it's consistent and meaningful.

Overall fine-tuning on the essay domain helped the model learn task-specific patterns, even though the base model had been pre-trained on much larger and more diverse datasets.

Qualitative Comparison:

Sample index: 234

Original essay:

I met my first semipalmated sandpiper in a crook of Jamaica Bay, an overlooked shore strewn with broken bottles and religious offerings at the edge of New York City. I didn't know what it was called, this small, dun-and-white bird running the flats like a wind-up toy, stopping to peck mud and racing to join another bird like itself, and then more. Soon a flock formed, several hundred fast-trotting feeders that at some secret signal took flight, wheeling with the flashing synchronisation that researchers observing starlings have mathematically likened to avalanche formation and liquids turning to gas. Entranced, I spent the afternoon watching them. The birds were too wary to approach, but if ...

Ground truth summary:

Animals have thoughts, feelings and personality. Why have we taken so long to catch up with animal consciousness?

Original model summary:

semipalmated sandpipers are the most common shorebird in North America. they eat molluscs, insect larvae and diatom algae growing in shallow, sun-heated

seawater. the semipalmated sandpipers are the most common shorebird in north America, with an estimated population around 1.9 million.

Fine-Tuned model summary:

The semipalmated sandpipers are the most common shorebird in North America, with an estimated population around 1.9 million

Comment:

Both focused on bird facts instead of the deeper theme about animal consciousness. The fine-tuned version was shorter but didn't really capture the philosophy of the original.

Sample index: 296

Original essay:

One late evening in the early summer of 1981, lying sleepless in my student bedsit at the top of a house in the Fallowfield district of Manchester, I became aware of a pattern of bright flashing lights on the wall. All I could see through the curtainless window on the opposite side of the room was a strip of rather cloudy night sky. The vivid flashing was coming from within, or perhaps behind, a bank of cloud. As I continued to watch, an object materialised from within the cloud, advancing until it stood in plain view in the night sky. It was a strikingly large craft of some kind, flattish but with rounded edges, like an old-fashioned bedwarmer, or perhaps a huge English muffin. It was spark ...

Ground truth summary:
UFO sightings are down. Ghosts are in decline. Are we more discerning now, or just afraid to trust anything?

Original model summary:
there was nothing remotely similar to my sighting, and there never were. there was nothing that resembled my sighting, and nothing at all in the whole of the UK for the month in question.

Fine-Tuned model summary:
Is there any UFOS, and there never were? It's not just flying saucers that have receded into history

Comment:

The fine-tuned model tried to sound more like a conclusion, while the base version focused on factual description. Again both missed the abstract reflection from the ground truth, but the fine-tuned version was slightly closer in tone.

Summary: Fine-tuning helped improve the structure and clarity of outputs but didn't help much with abstract understanding. The model learned patterns of essay summaries but still lacked true conceptual grasp.

PART-3:

I used Meta-Llama-3-8B-Instruct, loaded with 4-bit quantization on Colab. This setup ran on an NVIDIA A100 GPU, and I used essays 234 and 296 (same as in Part 2).

Prompts:

```
PROMPT_1 = """Summarize the following essay in one or two sentences.  
Essay:  
{essay}  
Summary: """ # Direct instruction
```

Reasoning: This is my baseline approach a simple, straightforward command with no extra guidance. I wanted to test how well the LLM performs with minimal instructions, relying only on its pre-trained understanding of what "summarize" means. The wording is plain to see if Llama 3's instruction-tuning is sufficient without any special prompting techniques. I expected this to produce accurate but possibly generic summaries.

```
PROMPT_2 = """You're writing a quick summary of this essay for someone who hasn't read it yet.  
Read the following essay and create a summary that:  
1. Gets the main point or argument across clearly  
2. Is about 1-2 sentences long  
3. Sounds natural and easy to understand  
4. Sticks to the big ideas, not small details  
Essay:  
{essay}  
Summary: """ # Structured requirements
```

Reasoning: I used a numbered list to give the model explicit, concrete guidelines about what makes a good summary. This approach is based on the idea that

breaking down the task into specific criteria helps the model understand exactly what I'm looking for. I included sounds natural because I noticed base models sometimes produce awkward phrasing, and "big ideas, not small details" to encourage abstract thinking. I expected this to produce more focused and well-structured summaries than Prompt 1.

```
PROMPT_3 = """Here are examples of essays and their summaries:
Example 1:
Essay: Online shopping has taken over everything. You can buy groceries,
-furniture, even cars without leaving your couch. It started with Amazon but,
-now every store has an app. The upside is you save time and sometimes money,
-with online deals. The downside is you can't actually see or touch what,
-you're buying, return shipping is a hassle, and a lot of people impulse-buy,
-stuff they don't need just because it's so easy. Plus all those delivery,
-trucks are bad for the environment.
Summary: Online shopping offers convenience and potential savings but,
-encourages impulse purchases, creates return complications, and contributes,
-to environmental concerns through excessive packaging and delivery emissions.

Example 2:
Essay: Coffee culture has gotten out of control lately. A regular coffee used,
-to cost like a dollar but now people are paying seven bucks for a fancy,
-latte with oat milk and special foam. Coffee shops are packed with people,
-working on laptops for hours, and there's a new trendy cafe opening every,
-week. Some people say it's about the experience and the Instagram photos,
-not just the caffeine. But it's making coffee less accessible for people who,
-just want a simple affordable drink without all the fancy extras.
Summary: The rise of specialty coffee culture has transformed coffee from an,
-affordable daily beverage into an expensive lifestyle product, prioritizing,
-aesthetic experience over accessibility and value.

Now, following the same style, summarize this essay:
Essay:
{essay}
Summary: """ # Few shot learning
```

Reasoning: I wanted to teach the model by showing it examples of the input-output pattern I expected. The two examples I chose demonstrate how to balance concrete details with abstract themes, which is exactly what the essay summaries in Part 2 required. I picked topics that are relatable and contemporary to match the style of the Aeon essays. By showing the model good summaries, I expected it to learn the right level of abstraction and the appropriate tone. This is a classic few shot learning approach, and I expected it to perform the best among all my prompts.

```
PROMPT_4 = """You're texting someone who asked what the essay was about, but,
-you're trying to keep it under 160 characters because you don't want to send,
-a long message. Give them the quick version.

Here's the essay:
{essay}
Your text message summary: """ # In-Context learning
```

Reasoning: I created a scenario-based prompt to give the model context about the communication situation rather than just asking for a summary. By framing it as "texting a friend" with a character limit, I hoped the model would naturally produce brief, casual, and conversational summaries. The 160-character constraint was meant to force extreme brevity. I expected this approach to produce the shortest and most casual summaries, though possibly at the cost of completeness and accuracy.

Comparison of Summaries:

Essay 1 (Sandpipers and consciousness)

Prompt	Summary
Ground truth	Animals have thoughts, feelings and personality. Why have we taken so long to catch up with animal consciousness?
Prompt 1 (Direct)	The author recounts their encounter with a flock of semipalmated sandpipers, describing their behavior and migration patterns.
Prompt 2 (Structured)	The essay describes the author's encounter with sandpipers in Jamaica Bay, highlighting their features and habits.
Prompt 3 (Few-Shot)	The author observes a flock of sandpipers and reflects on their synchronized movement and beauty. Emphasizes admiration for their unique traits.
Prompt 4 (In-Context)	"Essay's about a guy who finds a sandpiper in Jamaica Bay & watches them for hours, learning about their habits & unique features."

Comment:

Few-Shot produced the richest and most accurate summary overall. In-Context sounded casual and short fun to read but too simple.

Essay 2 (UFO Sightings)

Prompt	Summary
Ground truth	UFO sightings are down. Ghosts are in decline. Are we more discerning now, or just afraid to trust anything?
Prompt 1 (Direct)	The writer describes witnessing a UFO hovering in the night sky outside their window.
Prompt 2 (Structured)	The author describes seeing a silver object with flashing lights and reflects on how it challenged their sense of reality.
Prompt 3 (Few-Shot)	The author recalls a night in 1981 when they saw a UFO in the sky, describing it as a silver object that hovered, then disappeared. It left a lasting impression.
Prompt 4 (In-Context)	"Saw UFO in the sky outside my bedroom window in Manchester in 1981. Spooky!"

Comment:

The Few-Shot prompt was again the best clear, complete and close to the ground truth tone. In-Context was concise but too surface-level.

Few-Shot consistently gave the best summaries because it learned from examples. It picked up on the right level of detail and tone. In-Context was fun and super short but too casual for academic work. The texting constraint I added made it skip important themes. None of the prompts helped Llama capture the abstract philosophical points in the essays. Both essays had deeper meanings (animal consciousness, declining belief) but all four prompts just summarized the surface-level story. This shows LLMs still struggle with implicit themes no matter how we prompt them.

=====		
SUMMARY STATISTICS		
=====		
Prompt	Essay 1	Essay 2
=====		
Prompt 1 (Direct)	389	330
Prompt 2 (Structured)	408	381
Prompt 3 (Few-Shot)	544	575
Prompt 4 (In-Context)	152	94
Ground Truth: 113	108	

General Reflections:

Difficult Parts:

- Part 1 honestly took the most time almost one and half hour just to compute all the metrics and run the evaluation. The Perplexity part especially took a while because the model generated huge outputs and printing them slowed things down. I had to tweak the code to avoid printing everything and just show a few values (like the first 10), which made things manageable.
- Another tricky part was in Part 2, when I was setting up training I initially used `evaluation_strategy` which threw an error and I had to change it to `eval_strategy` for the trainer to work. Debugging that slowed me down a bit. Also, seeing lower perplexity on the base model but higher ROUGE on the fine-tuned model was confusing at first.
- For Part 3, setting up Meta's Llama-3 8B model with quantization worked fine, but I had to wait for access and approval from Hugging Face to use some models like the Llama base one. So I switched to Mistral-7B initially, and later came back to Llama-3 once I got access. That was kind of a blocker at first, but it got resolved by being patient.

What did you learn from this assignment?

This assignment gave me a solid idea of how summarization can be done in different ways:

- Fine-tuning helps with structure but not always meaning.
- Prompting can be more powerful than expected, especially Few-Shot prompts.
- Metrics like ROUGE and BERTScore tell part of the story but to really judge the summary, you have to read it yourself.

It also helped me understand how much tokenization, model type, and dataset can affect output quality.

How difficult did you feel this assignment was overall?

It was definitely one of the more involved assignments. I'd say a medium-hard in terms of difficulty not because of one super hard part, but because there were so many steps and it spanned multiple skills.

What would you do differently next time:

I will be intermediate outputs from the beginning to avoid recomputation. Explore prompts earlier to get deeper, more thoughtful summaries.